SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5111

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 | **Total Marks: 60**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I B. Shyamsai (19212174) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

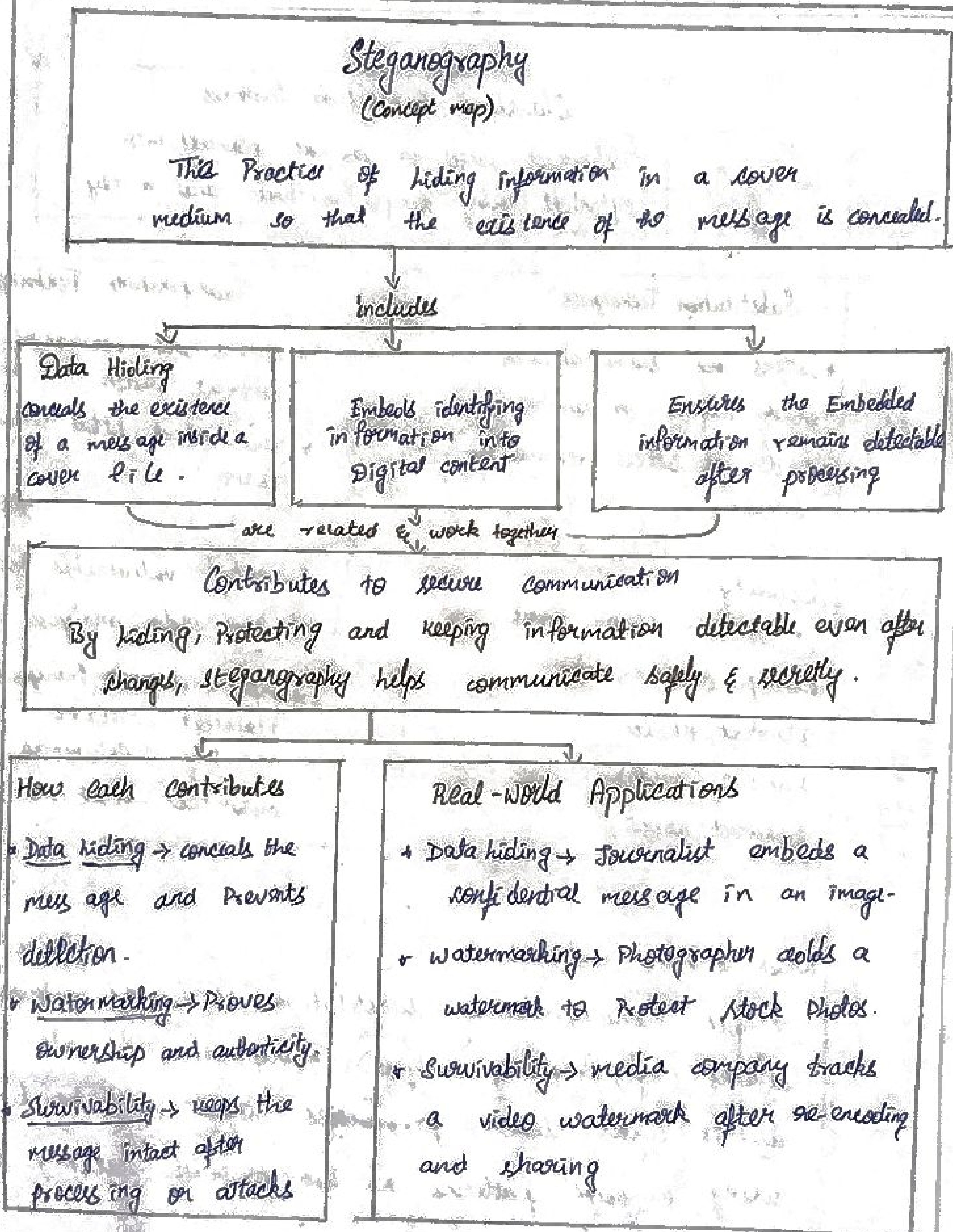
Classical Encryption Schemes

Techniques used to convert plaintext into ciphertext using simple methods and a key.

Substitution Techniques	Transposition Techniques
- letters are submitted with other letters or symbols. - order of letters remains the same	- letters are rearranged in a different order. - The actual letters remain the same
Security: - weak in simple forms.	Security: vulnerable to pattern and order analysis
Example (Caesar cipher) Plaintext : HELLO key : 3 (Ceasar) Ciphertext: KHOOR	Example (Columnar transposition) Plaintext : HELLO key : Columnar cipher text : HLEOL

Which is more vulnerable?

Simple monoalphabetic substitution is more vulnerable to attacks (eg., frequency analysis). Transposition is also week, as it leaks letter frequencies and can be broken using language patterns. Neither provides strong security when used alone.



Steganography

(Concept map)

The Practice of hiding information in a cover medium so that the existence of the message is concealed.

includes

Data Hiding
conceals the existence of a message inside a cover file.

Embeds identifying information into Digital content

Ensures the Embedded information remains detectable after processing

are related & work together

Contributes to secure communication

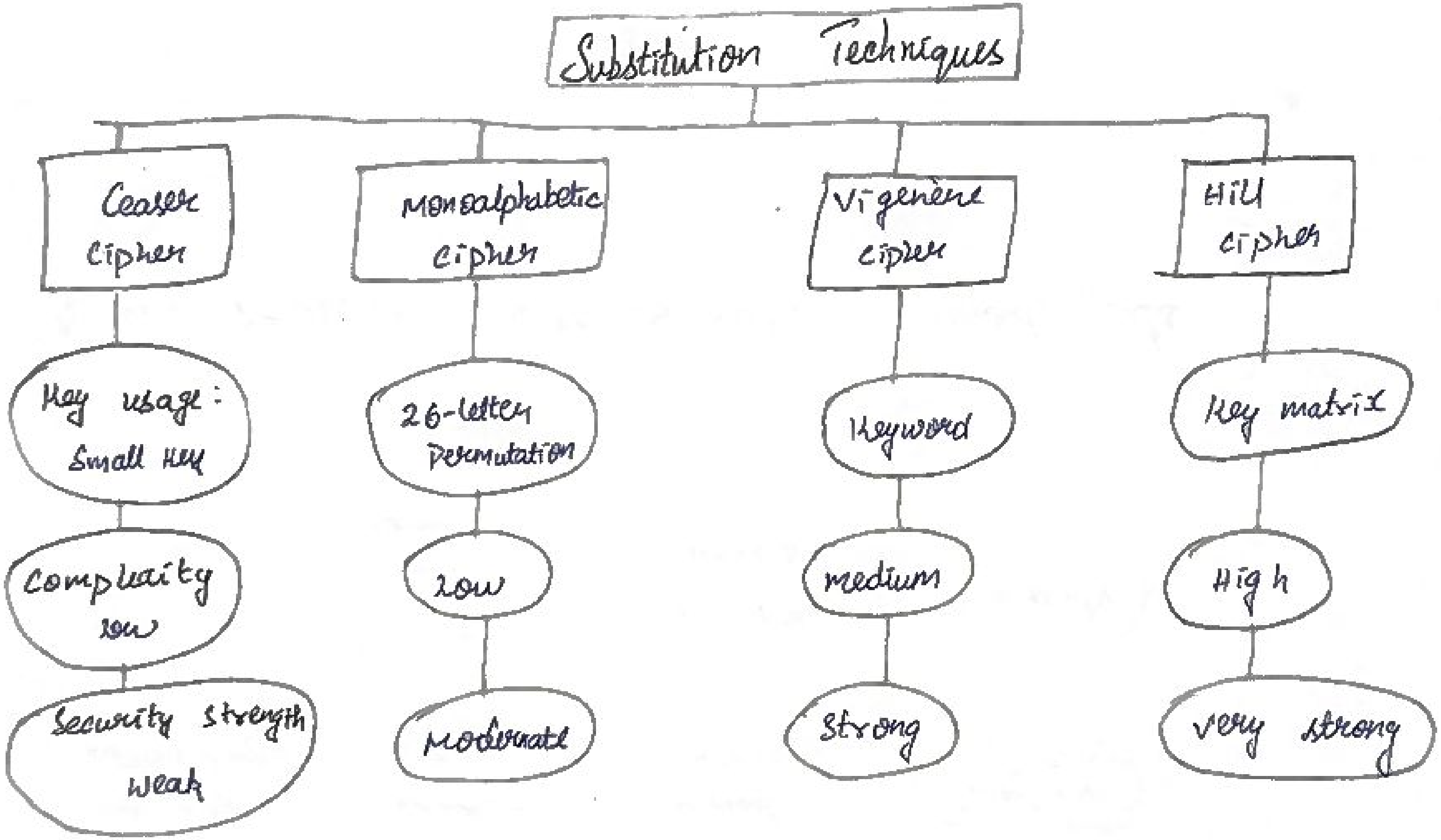
By hiding, Protecting and keeping information detectable even after changes, stegangraphy helps communicate safely & secretly.

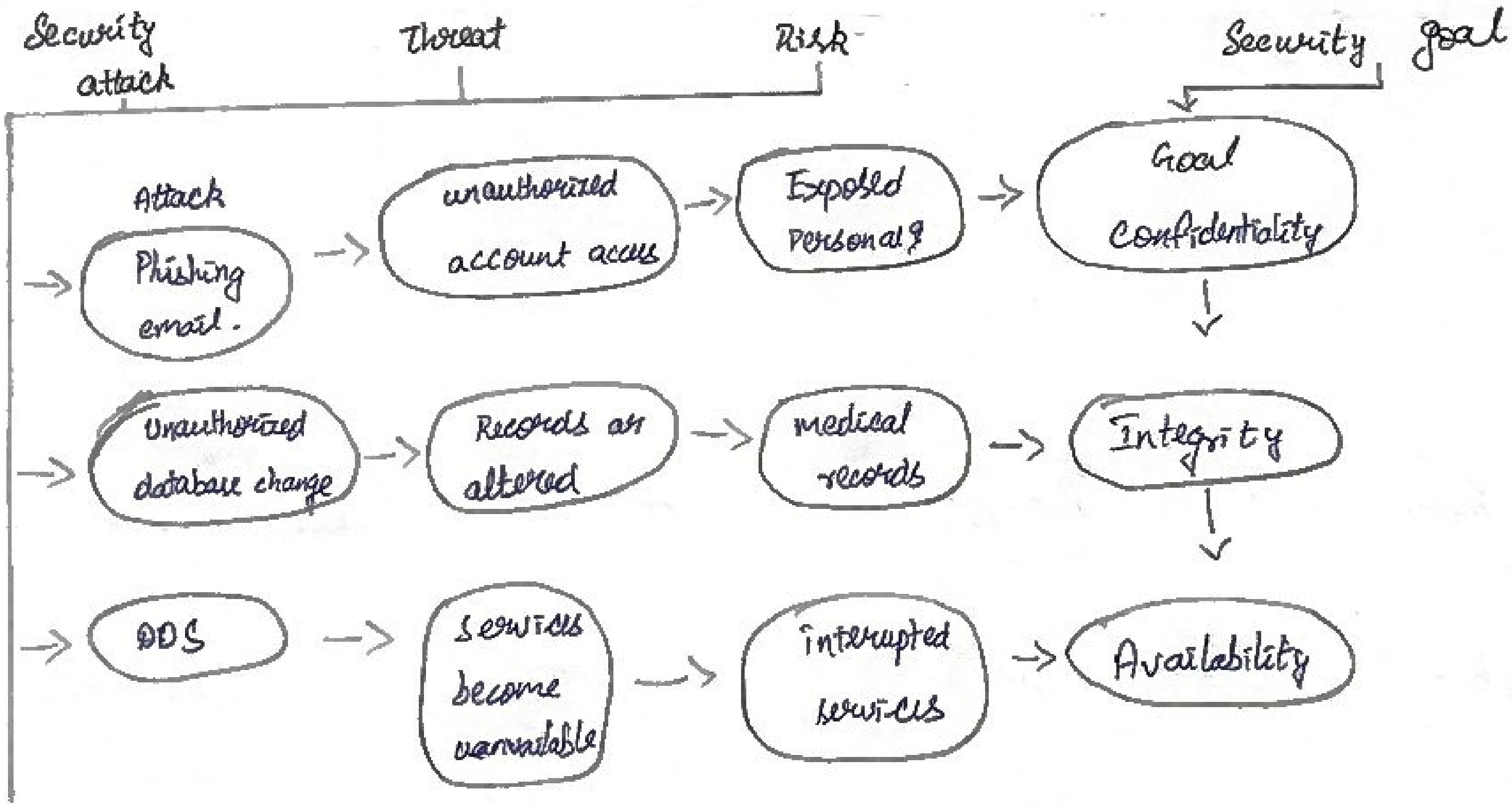
How each contributes

- Data hiding → conceals the message and Prevents detection.
- Watermarking → Proves ownership and authenticity.
- Survivability → keeps the message intact after processing or attacks

Real-world Applications

- Data hiding → Journalist embeds a confidential message in an image.
- watermarking → Photographer adds a watermark to Protect stock Photos.
- Survivability → media company tracks a video watermark after re-encoding and sharing


Substitution Techniques
Ceaser cipher
Monoalphabetic cipher
Vigenère cipher
Hill cipher
Key usage: Small key
26-letter permutation
Keyword
Key matrix
Complexity low
low
medium
High
Security strength weak
Moderate
Strong
very strong



Security Attacks → Threats → Risks → Security goals.